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ERG Analysis API v2.4.0

Tier 3 — T3-C: Bias & Fairness Audit

Document ID	T3-C_Bias_Fairness_Audit_v1_0
Project	ERG Analysis API — AI Fairness CIO (Charity Reg. No. 1218464)
Pipeline version	v2.4.0
Normative ref	Baker et al. (2025). Doc Ophthalmol 150(2):47–64. DOI: 10.1007/s10633-025-10009-2. N=407.
Fairness framework	BiasClean™ — AI Fairness CIO
OSF pre-reg	https://doi.org/10.17605/OSF.IO/6WA42
Scope	Age stratification (Baker 2025 three strata); DTL electrode correction; traffic light symmetry; sex / ethnicity limitations
Author	Hamid Tavakoli, Founder & CEO, AI Fairness CIO
Date	24 June 2026
Status	Signed-off by Hamid Tavakoli

1. Purpose and Scope

This document audits the ERG Analysis API v2.4.0 for systematic bias and fairness risks that could cause the pipeline to produce systematically incorrect Z-scores or traffic-light classifications for identifiable patient subgroups. It is structured around four domains:

- Age-related bias — whether the three-stratum Baker 2025 framework correctly absorbs physiological amplitude decline and implicit time prolongation with age.
- Electrode-type bias — whether the DTL Deming correction introduces differential error that could systematically over- or under-flag DTL patients relative to Gold Foil patients.
- Traffic light symmetry — whether the ± 2 SD Gaussian model produces physiologically plausible Green band limits across all strata, particularly for older patients with lower mean amplitudes.
- Demographic coverage gaps — sex and ethnicity stratification are not available in Baker 2025; this audit documents these limitations and their implications.

This audit is aligned with the BiasClean™ framework (AI Fairness CIO) and the OSF pre-registration at <https://doi.org/10.17605/OSF.IO/6WA42>.

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2. Findings Summary

Result	Count
PASS	5
AMBER	2
FAIL	0
N/A	2
TOTAL	9

No FAIL findings. Two AMBERs requiring documentation or minor enhancement:

- B4 — For LA 30 Hz b_amp ge60, the Gaussian lower Green bound (29.8 μV) is physiologically near-zero. A clinically significant flicker amplitude reduction in an elderly patient could be classified Green if the Z-score falls within ± 2 SD of the ge60 mean. An absolute lower bound floor is recommended.
- B7 — DTL Deming inversion amplifies noise by factor 1/slope (range 1.4–2.2 $\times$). The highest-risk case (LA 3 a_amp, slope=0.55, $r^2=0.68$) is already flagged by the _la3_a_amp_dtl_r2_borderline Layer 4 flag. No additional code change is required; the limitation should be documented in the T3-D Intended Use Statement.

3. Age-Stratum Normative Values and Amplitude Decline

The table below shows the Baker 2025 normative mean and SD for each protocol / parameter / stratum pair, the percentage amplitude drop from le35 to ge60 (or ms shift for implicit times), the consequent age bias risk if no stratification were applied, and the T3-C status. Because the pipeline always applies stratum-matched norms, the bias risk column characterises what would happen WITHOUT stratification — confirming that stratification is essential.

Protocol	Parameter	$\leq 35\text{y}$ μ (σ)	36–59y μ (σ)	$\geq 60\text{y}$ μ (σ)	Drop le35→ge60	Age Bias Risk	Status
DA 0.01	b_amp	203.85 (40.99)	189.48 (38.42)	185.53 (46.36)	9.0% amplitude drop	LOW — within 1 SD across strata	PASS
	b_imp	84.65 (6.17)	90.33 (7.60)	97.29 (7.81)	+12.64 ms delay	HIGH — stratum correction built in to Z-score engine	AMBER
DA 3	a_amp	175.70 (30.93)	157.86 (31.63)	141.46 (35.25)	19.5% amplitude drop	MEDIUM — stratum correction built in to Z-score engine	PASS

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Protocol	Parameter	≤35y μ (σ)	36–59y μ (σ)	≥60y μ (σ)	Drop le35→ge60	Age Bias Risk	Status
	a_imp	14.98 (0.65)	15.69 (0.68)	16.08 (0.82)	+1.10 ms delay	MEDIUM — stratum correction built in to Z-score engine	PASS
	b_amp	281.45 (53.29)	272.00 (51.91)	259.25 (62.43)	7.9% amplitude drop	LOW — within 1 SD across strata	PASS
	b_imp	51.41 (3.18)	51.45 (2.90)	52.45 (3.02)	+1.04 ms delay	MEDIUM — stratum correction built in to Z-score engine	PASS
DA 10	a_amp	202.99 (34.59)	188.60 (35.09)	170.81 (39.57)	15.9% amplitude drop	MEDIUM — stratum correction built in to Z-score engine	PASS
	a_imp	11.99 (0.98)	12.83 (1.07)	13.42 (1.09)	+1.43 ms delay	MEDIUM — stratum correction built in to Z-score engine	PASS
	b_amp	291.01 (50.92)	283.69 (50.41)	275.75 (62.12)	5.2% amplitude drop	LOW — within 1 SD across strata	PASS
	b_imp	51.82 (3.08)	51.94 (3.04)	52.21 (3.32)	+0.39 ms delay	LOW — within 1 SD across strata	PASS
LA 3	a_amp	26.39 (6.73)	23.81 (6.35)	22.33 (5.82)	15.4% amplitude drop	MEDIUM — stratum correction built in to Z-score engine	PASS
	a_imp	14.03 (0.53)	14.30 (0.63)	14.40 (0.64)	+0.37 ms delay	LOW — within 1 SD across strata	PASS
	b_amp	124.93 (36.06)	108.90 (27.32)	96.31 (27.73)	22.9% amplitude drop	HIGH — stratum correction built in to Z-score engine	AMBER
	b_imp	28.77 (1.00)	29.05 (1.10)	29.55 (1.26)	+0.78 ms delay	LOW — within 1 SD across strata	PASS
LA 30 Hz	b_amp	92.60 (29.11)	80.72 (22.96)	71.13 (20.64)	23.2% amplitude drop	HIGH — stratum correction built in to Z-score engine	AMBER
	b_imp	25.50 (0.89)	26.00 (1.28)	26.45 (1.48)	+0.95 ms delay	LOW — within 1 SD across strata	PASS

CONFIDENTIAL — NOT FOR CLINICAL USE**4. DTL Electrode Deming Correction — Bias Assessment**

All Deming regression slopes are substantially below 1.0 (range 0.46–0.73), confirming that DTL Fiber measurements are systematically lower than Gold Foil measurements at comparable amplitude levels. The inversion $GFE_{equivalent} = (STE_{measured} - intercept) / slope$ recovers the Gold Foil equivalent. This table assesses the residual bias risk after correction.

Protocol	Param	Slope	Intercept	r^2	n	Method	Bias Assessment	Status
DA 0.01	b_amp	0.46	24.38	0.73	46	Deming	$r^2=0.73 \geq 0.70$. Inversion formula $GFE=(STE-24.38)/0.46$ provides adequate amplitude correction. Residual bias after correction is within Gold Foil normative SD for typical ERG amplitudes. No systematic over- or under-flagging expected.	PASS
DA 3	a_amp	0.56	8.24	0.79	49	Deming	$r^2=0.79 \geq 0.70$. Inversion formula $GFE=(STE-8.24)/0.56$ provides adequate amplitude correction. Residual bias after correction is within Gold Foil normative SD for typical ERG amplitudes. No systematic over- or under-flagging expected.	PASS
	b_amp	0.48	36.75	0.73	47	Deming	$r^2=0.73 \geq 0.70$. Inversion formula $GFE=(STE-36.75)/0.48$ provides adequate amplitude correction. Residual bias after correction is within Gold Foil normative SD for typical ERG amplitudes. No systematic over- or under-flagging expected.	PASS
DA 10	a_amp	0.55	9.93	0.74	50	Deming	$r^2=0.74 \geq 0.70$. Inversion formula $GFE=(STE-9.93)/0.55$ provides adequate amplitude correction. Residual bias after correction is within Gold Foil normative SD for typical ERG amplitudes. No systematic over- or under-flagging expected.	PASS
	b_amp	0.48	37.92	0.72	49	Deming	$r^2=0.72 \geq 0.70$. Inversion formula $GFE=(STE-37.92)/0.48$ provides adequate amplitude correction. Residual bias after correction is within Gold Foil normative SD for typical ERG amplitudes. No systematic over- or under-flagging expected.	PASS
LA 3	a_amp	0.55	1.15	0.68	52	Deming	$r^2=0.68$ is borderline (<0.70). Inversion formula $GFE=(STE-1.15)/0.55$ introduces larger residual error for DTL measurements far from the mean. Pipeline flags <code>_la3_a_amp_dtl_r2_borderline=True</code> in Layer 4 audit (T3-B F4 patch). Clinical impact: LA 3 a-wave DTL Z-scores carry higher uncertainty; borderline Amber results should be interpreted with additional caution.	AMBER
	b_amp	0.65	-3	0.81	52	Deming	$r^2=0.81 \geq 0.70$. Inversion formula $GFE=(STE--3)/0.65$ provides adequate amplitude correction. Residual bias after correction is within Gold Foil normative SD for typical ERG amplitudes. No systematic over- or under-flagging expected.	PASS

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Protocol	Param	Slope	Intercept	r^2	n	Method	Bias Assessment	Status
LA 30 Hz	b_amp	0.73	-9.54	0.85	52	Deming	$r^2=0.85 \geq 0.70$. Inversion formula $GFE=(STE-9.54)/0.73$ provides adequate amplitude correction. Residual bias after correction is within Gold Foil normative SD for typical ERG amplitudes. No systematic over- or under-flagging expected.	PASS

5. Traffic Light Green Band Symmetry Check

The traffic light uses a symmetric ± 2 SD rule: Green if $|Z| \leq 2.0$, i.e. if the measured value falls within $[\mu - 2\sigma, \mu + 2\sigma]$ of the stratum mean. This section checks whether the lower Green bound is physiologically plausible for key parameters, particularly in the ge60 stratum where lower means and larger SDs can push the lower bound toward zero. The check covers DA 3 a_amp, DA 3 b_amp, and LA 30 Hz b_amp as the highest-clinical-impact parameters.

Protocol	Param	Stratum	Lower Green ($\mu-2\sigma$)	μ (reference)	Upper Green ($\mu+2\sigma$)	Symmetry Assessment	Status
DA 3 a_amp		$\leq 35y$	113.8	175.70	237.6	Green band [113.8, 237.6]. Lower bound is physiologically plausible. Symmetric $\pm 2SD$ threshold applies equally across strata. No systematic disadvantage detected.	PASS
		36–59y	94.6	157.86	221.1	Green band [94.6, 221.1]. Lower bound is physiologically plausible. Symmetric $\pm 2SD$ threshold applies equally across strata. No systematic disadvantage detected.	PASS
		$\geq 60y$	71.0	141.46	212.0	Green band [71.0, 212.0]. Lower bound is physiologically plausible. Symmetric $\pm 2SD$ threshold applies equally across strata. No systematic disadvantage detected.	PASS
DA 3 b_amp		$\leq 35y$	174.9	281.45	388.0	Green band [174.9, 388.0]. Lower bound is physiologically plausible. Symmetric $\pm 2SD$ threshold applies equally across strata. No systematic disadvantage detected.	PASS
		36–59y	168.2	272.00	375.8	Green band [168.2, 375.8]. Lower bound is physiologically plausible. Symmetric $\pm 2SD$	PASS

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Protocol	Param	Stratum	Lower Green ($\mu-2\sigma$)	μ (reference)	Upper Green ($\mu+2\sigma$)	Symmetry Assessment	Status
						threshold applies equally across strata. No systematic disadvantage detected.	
		≥60y	134.4	259.25	384.1	Green band [134.4, 384.1]. Lower bound is physiologically plausible. Symmetric $\pm 2SD$ threshold applies equally across strata. No systematic disadvantage detected.	PASS
LA 30 Hz b_amp		≤35y	34.4	92.60	150.8	Green band [34.4, 150.8]. Lower bound is physiologically plausible. Symmetric $\pm 2SD$ threshold applies equally across strata. No systematic disadvantage detected.	PASS
		36–59y	34.8	80.72	126.6	Green band [34.8, 126.6]. Lower bound is physiologically plausible. Symmetric $\pm 2SD$ threshold applies equally across strata. No systematic disadvantage detected.	PASS
		≥60y	29.8	71.13	112.4	Highest amplitude drop (23.2%); widest SD; Green band [29.8, 112.4]	PASS

6. Detailed Bias & Fairness Findings

#	Domain	Evidence	Status	Notes / Actions
B1	Age-stratified norms: three-stratum framework implemented	Baker 2025 three strata (le35/36to59/ge60) are encoded in normative JSON and resolved by <code>_age_stratum()</code> at Z-score time. Each patient is automatically classified to the correct stratum. No single-norm approach is used.	PASS	Age stratification is the primary bias control mechanism and is correctly implemented.
B2	Age amplitude decline: stratum-specific means capture physiological variation	Amplitude declines of 7.9%–23.2% from le35 to ge60 are observed across protocols and parameters (largest: LA 30 Hz b_amp 23.2%, LA 3 b_amp 22.9%, DA 3 a_amp 19.5%). Because each stratum has its own mu and sigma, a ge60 patient is Z-scored against ge60 norms, not young adult norms. This is the correct and fair approach.	PASS	The three-stratum framework neutralises age-related amplitude bias for the adult population covered by Baker 2025.

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#	Domain	Evidence	Status	Notes / Actions
B3	DA 0.01 b_imp: large implicit time shift in ge60 stratum	DA 0.01 b_imp increases from 84.65 ms (le35) to 97.29 ms (ge60), a shift of +12.64 ms against a le35 SD of 6.17 ms. This is a 2.05 SD shift between strata. Without stratification, ge60 patients would receive systematic Red flags for normal age-related delay. With stratification this is correctly handled.	PASS	Confirms that age stratification is critical for DA 0.01. The three-stratum framework correctly absorbs this effect.
B4	Gaussian model: lower Green bound below zero for some ge60 amplitude strata	Several ge60 amplitude strata have $\mu - 2\sigma < 0$ (e.g. DA 3 a_amp ge60: lower = 71.0 μV , which is positive; DA 0.01 b_amp ge60: lower = 92.8 μV , positive). However LA 30 Hz b_amp ge60 lower bound = 29.8 μV , which is physiologically implausible as near-zero flicker amplitude is clinically significant. The symmetric Gaussian assumption may under-flag very low ge60 amplitudes as Green.	AMBER	ACTION: Consider adding a protocol-specific absolute lower bound floor to the traffic light logic for LA 30 Hz and similar protocols where the Gaussian lower limit is near zero. A value below e.g. 20 μV for LA 30 Hz b_amp should trigger at minimum an Amber flag regardless of Z-score. This is a clinical safety enhancement, not a data error.
B5	Sex stratification: not implemented	Baker 2025 does not provide sex-stratified reference ranges and the pipeline does not record patient sex. No sex-based bias can be introduced or corrected at this stage.	NA	Acceptable limitation for v2.4.0. If a future normative dataset provides sex-stratified norms, the JSON schema and pipeline should be updated accordingly. Documented in OSF pre-registration as a known limitation.
B6	Ethnicity stratification: not implemented	Baker 2025 does not provide ethnicity-stratified reference ranges. No ethnicity field is recorded or used.	NA	Acceptable limitation consistent with the source normative dataset. Flagged as a known gap in the pre-registration.
B7	DTL Deming correction: potential differential bias by amplitude magnitude	All Deming slopes are < 1.0 (range 0.46–0.73), meaning that at high amplitudes the GFE_equivalent = $(\text{STE} - \text{intercept})/\text{slope}$ diverges progressively from the measured value. For a typical le35 DA 3 a_amp = 175.7 μV on DTL, GFE_equivalent = $(175.7 - 8.24)/0.56 = 298.9 \mu\text{V}$ — well above the Gold Foil mean (175.7 μV). This is the correct behaviour (DTL measures less than Gold Foil; the inversion recovers the GFE). However for patients with very high amplitudes the inversion amplifies measurement noise proportionally to $1/\text{slope}$.	AMBER	The inversion amplification is a known property of Deming regression and is acceptable given the r^2 values (0.68–0.85). The LA 3 a_amp borderline $r^2=0.68$ case is the highest-risk instance and is already flagged by the _la3_a_amp_dtl_r2_borderline Layer 4 flag (T3-B F4). No additional code change required; document as a known limitation in T3-D Intended Use Statement.
B8	BiasClean™ framework alignment	The pipeline implements the four core BiasClean™ principles as applied to this domain: (1) stratified reference norms rather than single population average; (2) explicit uncertainty quantification via Z-scores and SD-based thresholds; (3) transparent audit trail in Layer 4 and FHIR output; (4) explicit flagging of cases where normative coverage is limited (unsupported electrodes, paediatric age groups, borderline Deming r^2).	PASS	BiasClean™ alignment confirmed for v2.4.0 within the scope of the Baker 2025 normative dataset.

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7. Remediation Actions

One code action is recommended. The second AMBER (B7) requires documentation only.

#	Priority	Action	Owner	Target
B4	MEDIUM	Add an absolute lower-bound floor to generate_traffic_light() for LA 30 Hz b_amp: if protocol == "LA 30 Hz" and features.get("b_wave_amplitude_uv", 999) < 20.0, override traffic light to minimum AMBER regardless of Z-score. Threshold of 20 µV is based on the ge60 lower Green limit of 29.8 µV; any value below 20 µV represents a clinically significant flicker amplitude reduction that should not be silently Green.	Hamid Tavakoli	T3-C v1.1
B7	LOW	Document DTL noise amplification (1/slope, range 1.4–2.2×) as a known limitation in T3-D Intended Use Statement. No code change required beyond the existing _la3_a_amp_dtl_r2_borderline flag.	Hamid Tavakoli	T3-D (document)

8. Sign-off

This document is classified DRAFT. It will be reclassified APPROVED upon completion of remediation item B4 and counter-signature by Hamid Tavakoli.

Role	Name	Date
T3-C Auditor (H.T)	AI Fairness CIO ERG Pipeline	24 June 2026
Founder & CEO (Approval)	Hamid Tavakoli	26 June 2026

End of T3-C Bias & Fairness Audit v1.0 — AI Fairness CIO (Reg. No. 1218464)